DIPLOMA PROGRAMME PRACTICE · HL · SET A

# Chemistry / Paper 2

**Worked mark scheme · 90 marks · 150 minutes**

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 2 · QUESTION 1 [15]

## Following a metal through two oxidation states

**(a) Write the noble-gas electron configurations of M, $M^{2+}$ and $M^{3+}$. [3]**

1 M: $[Ar]3d^6 4s^2$.

1 $M^{2+}$: $[Ar]3d^6$, removing 4s first.

1 $M^{3+}$: $[Ar]3d^5$.

**(b) Explain why the aquated ions may absorb different visible wavelengths even though the metal is the same. [3]**

1 Ligands split the metal d orbitals into groups of different energy.

1 Absorption promotes an electron between available d levels, with $\Delta E = hf$.

1 Oxidation state changes metal–ligand interactions and the energy separation.

**(c) A particular absorption has wavelength 510 nm. Determine the energy absorbed per mole of photons. Use $h = 6.63\times10^{-34}$ J s, $c = 3.00\times10^{8}$ m $s^{-1}$ and $N_A = 6.02\times10^{23}$ $mol^{-1}$. [3]**

1 Convert λ to $5.10\times10^{-7}$ m.

1 E per photon $= hc/\lambda = 3.90\times10^{-19}$ J.

1 E per mole $= 2.35\times10^{2}$ kJ $mol^{-1}$.

**(d) The $M^{3+}$ fraction of a 1.60 mmol mixture is first reduced completely to $M^{2+}$ using 48.25 C, with 100% current efficiency. Use $Q = n(e^-)F$ and $F = 96500$ C $mol^{-1}$. Determine the original amount of each ion. [3]**

1 $n(e^-) = 48.25/96500 = 0.500$ mmol.

1 $M^{3+} + e^- \rightarrow M^{2+}$, so 0.500 mmol $M^{3+}$ initially.

1 $M^{2+}$ initially $= 1.60 - 0.500 = 1.10$ mmol.

**(e) A second analyst infers the original proportions from the mixed solution's colour alone. Evaluate the inference and propose a quantitative alternative. [3]**

1 Observed colour is the mixture's transmitted light and cannot separate overlapping absorptions by inspection.

1 Calibrate the pure ions at two wavelengths with different relative sensitivities.

1 Use the two absorbances in simultaneous additive concentration equations under the same ligand conditions.

PAPER 2 · QUESTION 2 [15]

# Closing a magnesium oxide cycle

**(a) Write equations, including states, for the second ionization and each of the two electron-gain steps. [3]**

1 $Mg^{+}(g) \rightarrow Mg^{2+}(g) + e^{-}$.

1 $O(g) + e^{-} \rightarrow O^{-}(g)$.

1 $O^{-}(g) + e^{-} \rightarrow O^{2-}(g)$.

**(b) Determine the enthalpy of lattice dissociation, making the direction of the cycle explicit. [3]**

1 Formation of the gaseous ions from the elements requires $148+738+1451+249-141+744=3189$ kJ $mol^{-1}$.

1 $3189 + \Delta H(\text{lattice formation}) = -602$, giving $-3791$ kJ $mol^{-1}$.

1 $MgO(s) \rightarrow Mg^{2+}(g) + O^{2-}(g)$: $+3791$ kJ $mol^{-1}$.

**(c) Explain why the second electron-gain step requires energy and why this does not prevent MgO from forming exothermically. [3]**

1 An incoming electron is repelled by the negative $O^{-}$ ion.

1 Energy is needed to overcome that repulsion in the gas phase.

1 The large negative lattice formation term more than offsets the total gaseous-ion cost.

**(d) Predict how the magnitude of the lattice dissociation enthalpy changes on replacing $Mg^{2+}$ with $Ca^{2+}$. Explain your reasoning. [3]**

1 The charge product is unchanged.

1 $Ca^{2+}$ is larger, increasing closest ion separation.

1 Attraction weakens; the positive dissociation enthalpy decreases.

**(e) An analyst uses +498 instead of +249 for the oxygen atomization step. Find the analyst's dissociation enthalpy and explain the error. [3]**

1 The gaseous-ion total rises by 249 kJ $mol^{-1}$.

1 The apparent dissociation enthalpy becomes $+4040$ kJ $mol^{-1}$.

1 Only half a mole of $O_2$ is dissociated per mole of MgO.

PAPER 2 · QUESTION 3 [15]

# Substitution and stereochemical evidence

**(a) State the empirical rate expression consistent with the data, give the units of k, and identify the bond broken in the slow step of the proposed unimolecular pathway. [3]**

1 Rate=k[2-bromobutane].

1 k has units $s^{-1}$.

1 C–Br breaks heterolytically in the rate-determining ionization.

**(b) Represent ionization and attack by water, describing the origin and destination of each electron-pair arrow. [3]**

1 Arrow from the C–Br bond to Br, giving $Br^-$ and $CH_3CH^+CH_2CH_3$.

1 A water oxygen lone pair points to the carbocation carbon.

1 Attack gives $CH_3CH(OH_2^+)CH_2CH_3$.

**(c) Complete the mechanism and explain the formation of both configurations. [3]**

1 Another water molecule accepts a proton; the oxonium O–H bond pair returns to O, giving butan-2-ol.

1 The approximately planar carbocation can be attacked from either face.

1 Attack on opposite faces gives the two enantiomers.

**(d) A sample contains 54.0% of one enantiomer and 46.0% of the other. Determine its enantiomeric excess and assess whether the data require a perfectly free carbocation. [3]**

1 Enantiomeric excess=54.0−46.0=8.0%.

1 A completely free planar intermediate in an achiral environment would give equal access to both faces.

1 The small excess is consistent with unequal access, for example shielding by a departing ion; it does not require a free intermediate.

**(e) Contrast this result with a single concerted backside displacement on the same enantiomer by $OH^-$. [3]**

1 Rate=k[substrate][$OH^-$] for elementary bimolecular substitution.

1 Bond formation and leaving-group departure occur in one step.

1 Backside attack gives inversion at the reacting carbon, rather than near-racemization.

PAPER 2 · QUESTION 4 [15]

# A shift reactor at 700 K

**(a) Calculate ΔG° at 700 K and deduce whether the standard-state forward reaction is product-favoured. [3]**

1 $\Delta S^\circ = -0.0420$ kJ $K^{-1}$ $mol^{-1}$.

1 $\Delta G^\circ = -41.0 - 700(-0.0420) = -11.6$ kJ $mol^{-1}$.

1 Negative ΔG° means K>1.

**(b) Calculate the equilibrium constant and state how changing the catalyst affects it. [3]**

1 $\ln K = 11600/(8.31 \times 700) = 1.994$.

1 K=7.35 (accept 7.3–7.4).

1 A catalyst changes approach rates but leaves K unchanged at fixed temperature.

**(c) Starting with 1.00 mol of each reactant and no products, calculate the equilibrium amount of $H_2$. [3]**

1 With extent x, amounts are 1−x, 1−x, x, x.

1 $K = x^2/(1-x)^2$; the common pressure or volume factors cancel.

1 $x = \sqrt{K}/(1+\sqrt{K}) = 0.731$ mol (approximately).

**(d) A different mixture at 700 K has Q=15. Calculate the actual ΔG and predict its initial direction of net change. [3]**

1 $\Delta G = \Delta G^\circ + RT \ln Q$.

1 $-11.6 + 0.00831 \times 700 \times \ln 15 = +4.15$ kJ $mol^{-1}$.

1 The reverse reaction initially predominates since Q>K.

**(e) Find the temperature for K=1 and explain why compressing the equilibrium gas mixture at fixed temperature does not change its composition in this model. [3]**

1 K=1 requires $\Delta G^\circ = 0$, so $T = \Delta H^\circ/\Delta S^\circ \approx 976$ K.

1 Compression multiplies all partial pressures by the same factor.

1 Equal gaseous stoichiometric totals on both sides leave Q unchanged at K.

PAPER 2 · QUESTION 5 [15]

# Iron content of a process sample

**(a) Construct the balanced overall ionic equation using half-equations. [3]**

1 $MnO_4^-+8H^++5e^-\rightarrow Mn^{2+}+4H_2O$.

1 $Fe^{2+}\rightarrow Fe^{3+}+e^-$.

1 $MnO_4^-+8H^++5Fe^{2+}\rightarrow Mn^{2+}+4H_2O+5Fe^{3+}$.

**(b) Calculate the mass percentage of iron in the original sample. [3]**

1 The aliquot consumes $4.000\times10^{-4}$ mol $MnO_4^-$, equivalent to 0.002000 mol $Fe^{2+}$.

1 The sample contains 0.02000 mol Fe, or 1.117 g.

1 Iron=89.36% (accept 89.4%).

**(c) Determine the standard cell potential and ΔG° per mole of permanganate consumed. Use F=96500 C $mol^{-1}$. [3]**

1 $E°cell=1.51-0.77=+0.74$ V.

1 Use $\Delta G°=-nFE°$, with n=5 per mole of permanganate.

1 $\Delta G°\approx-357$ kJ $mol^{-1}$.

**(d) A small fraction of $Fe^{2+}$ oxidizes in air during dilution. Predict the changes to titre and calculated percentage, and suggest a control. [3]**

1 Oxidized $Fe^{3+}$ does not consume permanganate as $Fe^{2+}$ does; the titre decreases.

1 The apparent Fe percentage is too low.

1 Prepare and titrate promptly under oxygen-limiting conditions, e.g. deoxygenated solution under inert gas.

**(e) Each burette reading has uncertainty ±0.05 $cm^3$. Estimate the maximum percentage uncertainty from the titre and distinguish it from the oxidation error. [3]**

1 Maximum absolute uncertainty in the difference=0.10 $cm^3$.

1 Percentage=100×0.10/20.00=0.50%.

1 Reading uncertainty can affect measurements in either direction; prior oxidation biases the result systematically low.

PAPER 2 · QUESTION 6 [15]

# Identifying a volatile carbonyl

**(a) Calculate the index of hydrogen deficiency and explain which feature accounts for it. [3]**

1 IHD=(2×5+2−10)/2=1.

1 The IR band is consistent with C=O.

1 One carbonyl double bond uses the single degree of unsaturation, leaving no ring or other double bond.

**(b) Deduce the structure and assign both signals, including the reason for each splitting pattern. [3]**

1 Pentan-3-one: $CH_3CH_2C(=O)CH_2CH_3$.

1 Two equivalent terminal methyl groups couple to adjacent $CH_2$ (two neighbours), giving the 6H triplet.

1 Two equivalent $CH_2$ groups each couple to three methyl protons, giving the 4H quartet.

**(c) Explain why pentan-2-one and pentanal each fail at least one observation and why the molecular-ion peak alone cannot distinguish them. [3]**

1 Pentan-2-one has an acetyl methyl singlet and lacks equivalence of two ethyl groups.

1 Pentanal would show an aldehydic proton signal and additional environments.

1 All three share the same molecular formula and nominal molecular mass.

**(d) Predict the main product of reduction of the carbonyl by a suitable hydride reagent followed by protonation. Discuss chirality. [3]**

1 Pentan-3-ol, $CH_3CH_2CH(OH)CH_2CH_3$.

1 The former carbonyl carbon is tetrahedral after reduction.

1 It carries two identical ethyl groups so is not stereogenic.

**(e) An unknown is proposed to be a mixture of unchanged ketone and its alcohol. Give two independent spectral observations expected for this mixture and explain why a broad O–H peak alone is insufficient. [3]**

1 Residual C=O absorption supports remaining ketone.

1 New alcohol C–H proton environment on CH(OH) in $^1$H NMR (or a distinct alcohol molecular ion at 88).

1 Water or another O–H-containing impurity could also cause broad O–H absorption.